

Poisson distribution

Poisson distribution

$$P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!} \quad x=0, 1, 2, \dots$$

$$\begin{aligned}
 E(x) &= \sum_{k=0}^{\infty} x^k \frac{1}{k!} (x = \lambda) \\
 &= \sum_{k=0}^{\infty} x^k \frac{\lambda^k}{k!} \\
 &= \left[1 \cdot e^{\lambda} \frac{\lambda^0}{0!} + 2 \cdot e^{\lambda} \frac{\lambda^1}{1!} + 3 \cdot e^{\lambda} \frac{\lambda^2}{2!} + \dots \right] \\
 &= \lambda e^{\lambda} \left[1 + \frac{\lambda}{1!} + \frac{\lambda^2}{2!} + \dots \right] \\
 &= \lambda e^{\lambda} e^{\lambda} \\
 &= \lambda e^{2\lambda}
 \end{aligned}$$

$$\begin{aligned} \text{Var}(X) &= E(X^2) - (E(X))^2 \\ &= E(X(X-1)) + E(X) - (E(X))^2 \end{aligned}$$

$$\begin{aligned} E(X(X-1)) &= \sum_{n=0}^{\infty} n(n-1) \frac{e^{-\lambda} \lambda^n}{n!} \\ &= 2 e^{-\lambda} \frac{\lambda^2}{2!} + 3 \cdot 2 e^{-\lambda} \frac{\lambda^3}{3!} + 4 \cdot 3 e^{-\lambda} \frac{\lambda^4}{4!} + \dots \\ &= e^{-\lambda} \lambda^2 \left[1 + \frac{\lambda}{1!} + \frac{\lambda^2}{2!} + \dots \right] \\ &= e^{-\lambda} \lambda^2 e^{\lambda} \\ &= \lambda^2 \end{aligned}$$

$$\text{Var}(X) = 1^2 + 1 - 1^2 = 1$$

MGF:

$$\begin{aligned}
 MGF: \\
 M(t) &= E(e^{tx}) \\
 &= \sum_{n=0}^{\infty} \frac{t^n e^{-\lambda} \lambda^n}{n!} \\
 &= \sum_{n=0}^{\infty} \frac{(\lambda t)^n}{n!} e^{-\lambda} \\
 &= e^{-\lambda} \sum_{n=0}^{\infty} \frac{(\lambda t)^n}{n!} \\
 &= e^{-\lambda} e^{\lambda t} = e^{\lambda(t-1)}
 \end{aligned}$$

Eg The average number of house sold by
a company is 2 homes per day.
. " . " that exactly 3 homes

Eg The average ...
 a company is 2 homes per day.
 What is the prob that exactly 3 homes
 will be sold on a day?

Sol: X : number of house sold by that company
 per day. $X \sim P(\lambda)$

$$E(X) = 2$$

$$\lambda = 2$$

$$P(X=3) = \frac{e^{-2} \frac{2^3}{3!}}{1} = 0.1804$$

Eg Suppose the average number of lions
 seen per day in a jungle safari
 is 5. What is the probability that
 the tourist will see less than 4 lion
 on a day safari.

$$X \sim P(5)$$

$$P(X < 4) = \sum_{x=0}^3 P(X=x)$$

Eg Cars pass through a junction on a busy
 road at an average rate of
 300 per hour.
 find the prob that ~~no~~ no car passes a
 given minute.

Sol: X : # cars passing per minute

$$X \sim P(\lambda)$$

$$\lambda = \frac{300}{60} = 5$$

$$P(X=0) = \frac{e^{-5} 5^0}{0!} = e^{-5}$$

→
 ... is a special case of Binomial distribution

Poisson approximation to Binomial distribution

$$X \sim \text{Bin}(n, p)$$

- (i) the number trials $n \rightarrow \infty$
- (ii) success prob $p \rightarrow 0$
- (iii) $np \rightarrow \lambda (> 0)$ finite

$$X \sim P(\lambda)$$

X : # accidents / day.

$$n = \frac{1}{100}$$

$$p \rightarrow 0$$

$$np = \lambda$$

$$np = \lambda$$

$$\text{Bin}(n, p)$$

$$n, \quad p = \frac{2}{n}$$

$$n \rightarrow \infty \quad p \rightarrow 0$$

$$(np \rightarrow 2)$$